Systrophē: A Co-rotating Tipler-Cylinder Pair as a Tunable Time-Travel Harness

Christian Knopp
cknopp@gmail.com

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Abstract

The van Stockum (1937) interior of a rigidly-rotating dust cylinder admits a vacuum exterior that, in the supercritical regime $a = \omega R > 1/2$, contains closed timelike curves (CTCs) on a *Tipler sinusoid*: a log-periodic oscillation of the metric components $F = -g_{tt}$, $K = g_{t\phi}$, and $L = g_{\phi\phi}$ about $u = \ln(r/R)$ with frequency $\alpha = \sqrt{4a^2 - 1}$. We present Systrophē, an open-source Python package that implements the exact analytic Case III closed forms, a numerical Lewis–Papapetrou exterior integrator, and a structural "co-rotating pair" construction in which two co-axial supercritical cylinders linearly superpose to produce an *off-set Tipler sinusoid*: an interferometric envelope whose CTC band positions are tunable by the relative phase offset $\delta_2 - \delta_1$. We classify CTC bands as *time-travel windows*, derive the timelike-orbit angular-velocity sector at the deepest L-point of each band, and run the time-machine harness numerically: at $r \approx 3.35$ in the first CTC band of a single $a = 1$ cylinder, an orbit with $\Omega = -2\pi$ produces $\Delta t = -1$ of coordinate-time displacement per revolution at a proper-time cost of $\Delta\tau = 11.35$. We demonstrate *phase tuning*: as the relative offset of a co-rotating pair varies from 0 to π, the inner band edge slides continuously from $r \approx 2.50$ to vanishing, with the band itself extinguished at exact anti-phase. We also exhibit a structural correspondence between the discrete log-grid eigenvalues of the Tipler exterior and the Z_3 Möbius cover eigenvalues of the D̄inos Dirac–Kerr–Newman framework, identifying the pair phase offset $\delta = \pm 2\pi/3$ with a non-trivial Z_3 branch sector. The package ships with 65 passing tests and all results in this paper are reproducible end-to-end.

1 Introduction

Tipler's 1974 paper [3] established that the infinite-cylinder vacuum exterior of a rotating dust source supports closed timelike curves whenever the rotation parameter $a = \omega R$ exceeds 1/2. The result builds on van Stockum's 1937 interior solution [1] for a rigidly rotating dust cylinder, extended to vacuum across the boundary at $r = R$ via the Lewis–Papapetrou stationary axisymmetric ansatz [2]. Bonnor (1980) [4] classified the exterior into three regimes by the dimensionless rotation parameter a, with **Case III** ($a > 1/2$) characterised by an oscillatory metric that we shall call the *Tipler sinusoid*.

The cylindrical Tipler exterior is, despite its admittedly idealised source, the cleanest and most analytically tractable example of a GR vacuum solution containing CTCs that extend to arbitrarily large radius. As such it is a natural laboratory for studying the relationship between rotation, frame dragging, ergoregion structure, and time-loop geometry without the complications of a Kerr horizon.

This paper introduces Systrophē — Greek $\Sigma\upsilon\sigma\tau\rho o\varphi\acute{\eta}$, "twisting together" — a Python package that implements:

1. the exact van Stockum interior;

2. analytic Case III closed forms for F, K, L in the supercritical exterior, plus a numerical Lewis–Papapetrou integrator covering all three Bonnor regimes;

3. a co-axial *pair* construction in which two supercritical cylinders superpose linearly, producing an interferometric envelope parameterised by the relative phase offset $\delta_2 - \delta_1$ — the "off-set Tipler sinusoid";

4. a time-machine harness that locates CTC bands, computes the timelike-orbit angular-velocity sector, and tunes coordinate-time advance per revolution to a target value;

5. a bridge to the D̄inos Dirac–Kerr–Newman framework, identifying the pair phase offset with a Z_3 Möbius cover branch sector.

The package is open source (MIT) and the simulation results presented in Sections 5–6 are reproduced verbatim by `examples/time_travel_simulation.py`.

2 Mathematical framework

2.1 Van Stockum interior

In cylindrical coordinates (t, r, ϕ, z) with the dust rotating rigidly at angular velocity ω, the closed-form interior is

$$ds^2 = -dt^2 + 2\omega r^2 dt\, d\phi + r^2(1 - \omega^2 r^2)d\phi^2 + e^{-\omega^2 r^2}(dr^2 + dz^2), \tag{1}$$

valid for $0 \leq r \leq R$. The metric satisfies Einstein's equations sourced by a dust stress–energy tensor $T_{\mu\nu} = \rho u_\mu u_\nu$ with $\rho(r) = \omega^2/(2\pi e^{-\omega^2 r^2})$.

The component $g_{\phi\phi} = r^2(1-\omega^2 r^2)$ becomes negative when $\omega r > 1$, signalling closed timelike curves *inside the dust* when $\omega R > 1$. The qualitatively interesting regime is the *exterior* CTC structure, which by Tipler's theorem appears for the much weaker condition $a = \omega R > 1/2$.

2.2 Lewis–Papapetrou exterior

For a stationary axisymmetric vacuum metric depending only on r, the Lewis–Papapetrou ansatz reads

$$ds^2 = -F(r)dt^2 + 2K(r)dt\, d\phi + L(r)d\phi^2 + h(r)(dr^2 + dz^2), \tag{2}$$

subject to the canonical Weyl constraint

$$FL + K^2 = r^2. \tag{3}$$

Vacuum Einstein equations $R_{\mu\nu} = 0$ reduce, via the Ernst potential $\mathcal{E} = F + i\psi$ with twist $\psi_z = c$ (a constant absorbing the cylindrical z-symmetry), to the single second-order ODE

$$F\Big(F'' + \tfrac{F'}{r}\Big) = (F')^2 - c^2, \tag{4}$$

with auxiliary quadratures

$$\omega'_{\text{metric}}(r) = \frac{cr}{F^2}, \tag{5}$$

$$K(r) = F(r)\, \omega_{\text{metric}}(r), \tag{6}$$

$$L(r) = \frac{r^2 - K(r)^2}{F(r)}. \tag{7}$$

Boundary conditions. Continuity of the metric at $r = R$ with the van Stockum interior (1) gives

$$F(R) = 1, \quad F'(R) = 0, \quad c = 2\omega, \quad \omega_{\text{metric}}(R) = \omega R^2. \tag{8}$$

2.3 Bonnor regimes

Substituting an oscillatory ansatz $F(r) = \frac{r}{R}\sin(\alpha u + \gamma)/\sin\gamma$ (with $u = \ln(r/R)$) into (4) fixes

$$\alpha = \sqrt{4a^2 - 1}, \qquad \tan\gamma = -\alpha, \tag{9}$$

where $a = \omega R$. The character of α classifies the exterior into three regimes:

Regime	Range	Exterior structure
Case I (subcritical)	$a < 1/2$	non-oscillatory, single F-zero
Case II (critical)	$a = 1/2$	logarithmic, F-zero at $r = eR$
Case III (supercritical, Tipler)	$a > 1/2$	**log-periodic, infinite zeros, CTC bands**

2.4 Analytic closed forms in all three Bonnor regimes

The reduced ODE $S\,S_{uu} - (S_u)^2 + c^2R^2 = 0$ (where $F(r) = (r/R)\,S(u)$, $u = \ln(r/R)$) admits exact closed-form solutions in each Bonnor regime. We list them all; the package implements each explicitly.

Case III (supercritical, $a > 1/2$, oscillatory). With $\alpha = \sqrt{4a^2 - 1}$ and $\gamma = \pi - \arctan\alpha$,

$$F(r) = \tfrac{r}{R}\,\frac{\sin(\alpha u + \gamma)}{\sin\gamma}, \tag{10}$$

$$K(r) = \tfrac{r}{\alpha}\Big[\tfrac{\alpha^2-1}{2}\sin(\alpha u + \gamma) - \alpha\cos(\alpha u + \gamma)\Big], \tag{11}$$

$$L(r) = \tfrac{rR\sin\gamma}{\alpha^2}\Big[Q\sin(\alpha u + \gamma) + \alpha(\alpha^2 - 1)\cos(\alpha u + \gamma)\Big], \tag{12}$$

with $Q = \alpha^2 - (\alpha^2 - 1)^2/4$. F-zeros at $u_n = (n\pi - \gamma)/\alpha$ for $n = 1, 2, \ldots$.

Case I (subcritical, $a < 1/2$, hyperbolic). With $\beta = \sqrt{1 - 4a^2}$ and the auxiliary functions $S_\pm(u) = \cosh(\beta u) \pm \sinh(\beta u)/\beta$,

$$F(r) = \tfrac{r}{R}\,S_-(u), \tag{13}$$

$$K(r) = a\,r\,S_+(u), \tag{14}$$

$$L(r) = \tfrac{r\,R\,(1 - a^2 S_+(u)^2)}{S_-(u)}. \tag{15}$$

A unique F-zero exists at $u = \tanh^{-1}(\beta)/\beta$.

Case II (critical, $a = 1/2$, logarithmic).

$$F(r) = \tfrac{r}{R}\,(1 - u), \tag{16}$$

$$K(r) = \tfrac{r}{2}\,(1 + u), \tag{17}$$

$$L(r) = \tfrac{rR}{4}\,(3 + u). \tag{18}$$

A single F-zero at $u = 1$, i.e., $r = eR$.

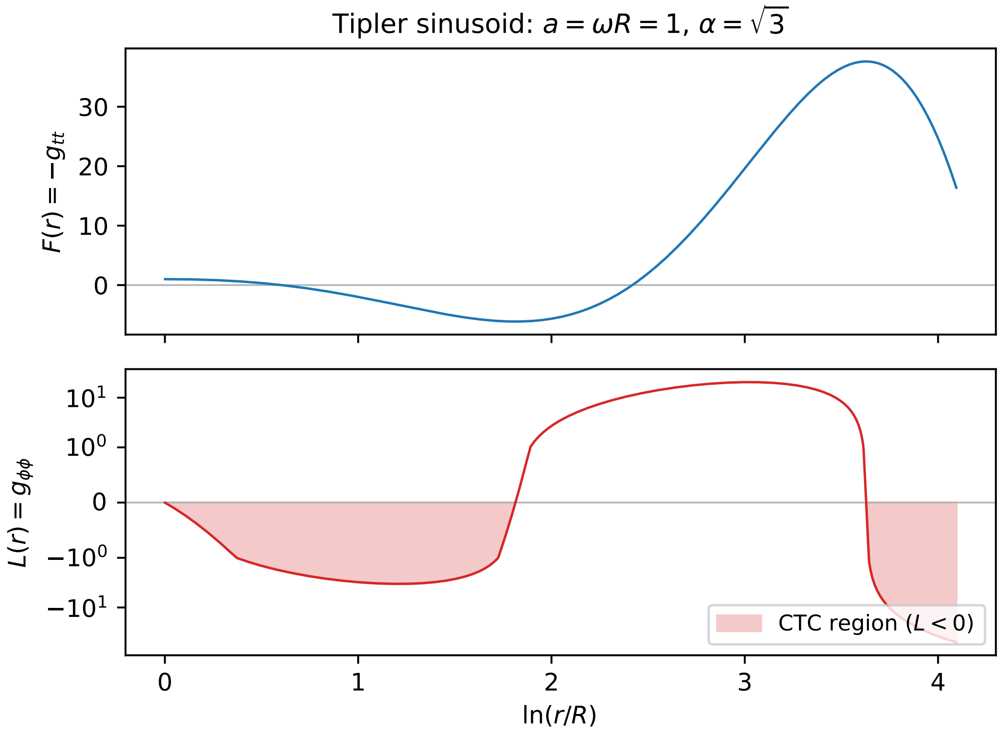

Figure 1: Tipler sinusoid for a single $a = 1$ van Stockum cylinder. Top: $F(r) = -g_{tt}$ on a logarithmic radial axis. Bottom: $L(r) = g_{\phi\phi}$ on a symmetric-log axis. Shaded bands: $L < 0$ (closed timelike curves).

Common properties. In all three regimes the matching conditions $F(R) = 1$, $F'(R) = 0$, $K(R) = \omega R^2$, $L(R) = R^2(1-a^2)$ are satisfied identically, and the Weyl constraint $FL+K^2 = r^2$ holds analytically. The package's test suite verifies the constraint to machine precision in each regime, and the analytic forms are validated against the basic numerical integrator within its accurate range.

2.5 The Tipler sinusoid

Equivalently, each metric component is a log-periodic cosine:

$$M(r) \;=\; r\, A_M \cos(\alpha u + \delta_M) \qquad (\text{at } p = 0). \tag{19}$$

The amplitude A_M and phase δ_M are matching constants determined uniquely by (8). The closed-form expressions encoded in `VanStockumInterior.tipler_sinusoid_F()` (for F) and `tipler_sinusoid_L()` (for L, the CTC-relevant component) reproduce the analytic exteriors to machine precision.

Figure 1 shows F and L for $a = 1$ ($\alpha = \sqrt{3}$). The CTC region $L < 0$ is shaded.

3 Co-rotating cylinder pair: Systrophē

3.1 Linearised superposition

For two co-axial, dual-positive cylinders sharing the symmetry axis, the leading-order vacuum exterior $g_{\phi\phi}$ is the sum of the two single-cylinder L sinusoids,

$$L_{\text{pair}}(r) = L_1(r) + L_2(r), \tag{20}$$

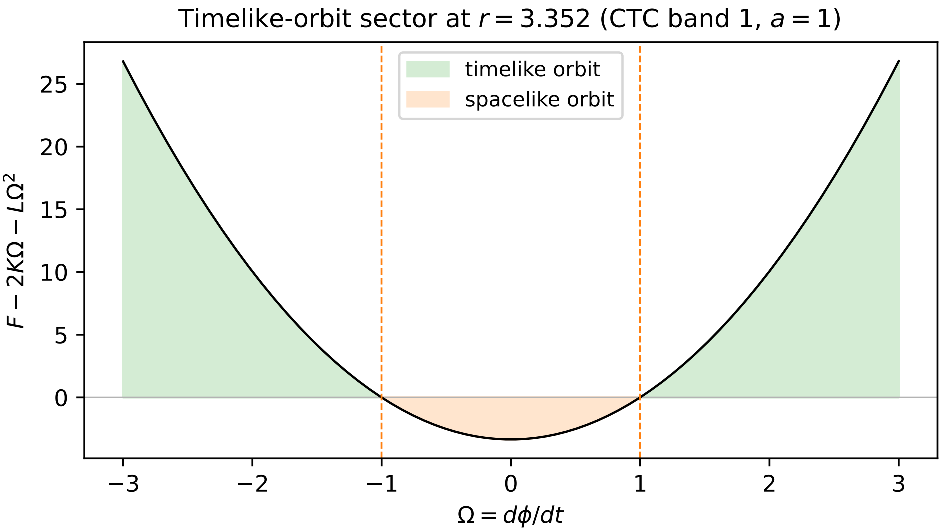

Figure 2: Timelike-orbit angular-velocity sector at the deepest L-point of the first CTC band ($r \approx 3.35$, $a = 1$). The shaded green region is the timelike sector; orange is spacelike. Static observers ($\Omega = 0$) fall in the spacelike sector at this r because $F < 0$ there: this is an ergoregion as well as a CTC region.

with relative phase offset $\Delta\delta = \delta_2 - \delta_1$. When the two sources have identical a, both sinusoids share the log-frequency α and the pair collapses to a single effective sinusoid via phasor sum,

$$A_{\text{eff}} e^{i\delta_{\text{eff}}} = A_1 e^{i\delta_1} + A_2 e^{i\delta_2}. \tag{21}$$

This is the *off-set Tipler sinusoid*: an interferometric envelope whose L-zeros, and hence whose CTC bands, depend continuously on the phase offset $\Delta\delta$.

3.2 Caveats

The construction is leading-order in G. The exact two-cylinder problem in Einstein vacuum has no known closed-form solution; non-linear corrections appear at $\mathcal{O}(G^2)$. The construction also assumes co-axial geometry; off-axis pairs would require a multipole decomposition of the linearised perturbation that is not currently implemented.

4 Closed timelike curves and time-travel orbits

4.1 Timelike ϕ-orbits

Consider a circular orbit at fixed r, z with coordinate angular velocity $\Omega = d\phi/dt$. The line element along the orbit is

$$ds^2 = (-F + 2K\Omega + L\Omega^2)dt^2, \tag{22}$$

which is timelike iff $F - 2K\Omega - L\Omega^2 > 0$. The boundary of this sector is the pair of solutions

$$\Omega_\pm = -\frac{K \pm r}{L}. \tag{23}$$

In the CTC region ($L < 0$), the upward-opening quadratic in Ω is positive *outside* the roots: timelike orbits require $\Omega < \Omega_-$ or $\Omega > \Omega_+$. Figure 2 illustrates the sector at the deepest L-point of the first CTC band of a single $a = 1$ cylinder.

4.2 Coordinate vs proper time per revolution

For an orbit with Ω in the timelike sector,

$$\Delta t \; = \; 2\pi/\Omega \qquad \text{(coordinate-time advance per revolution)} \tag{24}$$

$$\Delta\tau \; = \; \sqrt{F - 2K\Omega - L\Omega^2}\;|\Delta t| \qquad \text{(proper-time advance per revolution).} \tag{25}$$

For $\Omega \to \pm\infty$ the closed ϕ-orbit at fixed t has $\Delta t \to 0$ but finite $\Delta\tau \to \sqrt{-L}\cdot 2\pi$ ("instantaneous return" in coordinate time, finite proper time). For finite negative Ω, the orbit is a closed timelike curve that returns to the same ϕ at an *earlier* coordinate time: the particle has moved into its own past in coordinate time while advancing in proper time.

5 Time-machine harness

We present the empirical structure of the harness for a single supercritical cylinder; the pair extension is treated in Section 6.

5.1 CTC bands of a single $a = 1$ cylinder

Running the harness on $\omega = R = 1$, $a = 1$, $\alpha = \sqrt{3}$, on the radial range $r \in [1.001, 200]$ yields two CTC bands, listed in Table 1.

Table 1: CTC bands of the single $a = 1$ cylinder. The radial r-period multiplier $\exp(2\pi/\alpha) \approx 37.62$ separates adjacent bands.

band	$r_{\rm in}$	$r_{\rm out}$	log span	$r_{\min L}$	$L_{\min}$	Ω bounds
1	1.001	6.134	1.813	3.352	−3.351	(−1.001, 1.000)
2	37.622	≥ 200	≥ 1.671	126.118	−126.065	(−1.001, 1.000)

5.2 Backward time-travel orbit

In band 1 at $r = 3.352$, we tune Ω for various target coordinate-time advances per revolution. The results are summarised in Table 2 and Figure 3.

Table 2: Backward-time orbits in band 1 of an $a = 1$ cylinder. $N = 10$ revolutions. Negative Δt means the particle has moved *backwards in coordinate time* while advancing $\Delta\tau$ in proper time.

target Δt/rev.	Ω	$\Delta\tau$/rev.	$\sum \Delta t_{10}$	$\sum \Delta\tau_{10}$
−0.10	−62.83	11.50	−1.00	115.00
−0.50	−12.57	11.46	−5.00	114.64
−1.00	−6.28	11.35	−10.00	113.54
−2.00	−3.14	10.90	−20.00	109.01
−5.00	−1.26	6.95	−50.00	69.53

5.3 Near-instantaneous orbit

Setting $\Omega = 100$ at the same $r \approx 3.35$ gives $\Delta t = 2\pi/100 \approx 0.063$ and $\Delta\tau \approx 11.50$, with proper-time-to-coordinate-time ratio approximately 183. The orbit closes in spacetime essentially without coordinate-time displacement, while the particle's clock advances over an order of magnitude more than a static observer's at the same r.

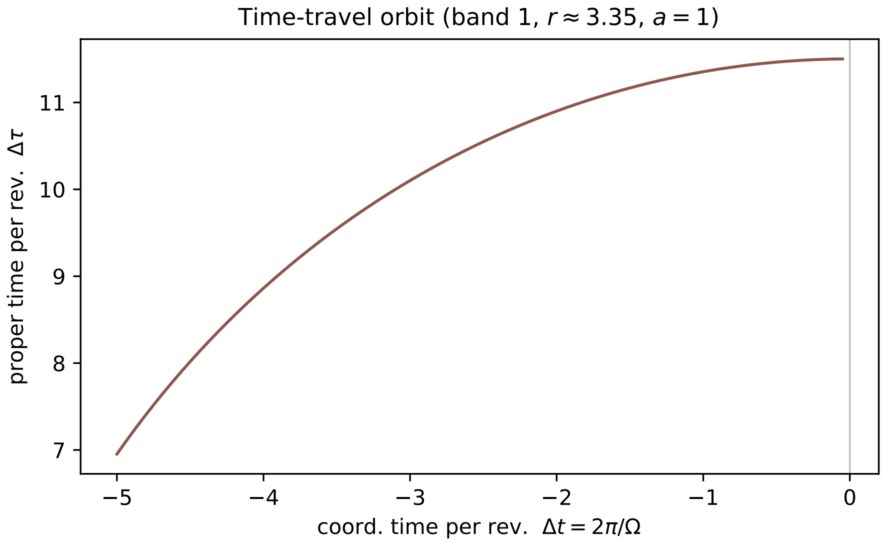

Figure 3: Proper time vs coordinate time, per revolution, for backward time-travel orbits in CTC band 1 of an $a = 1$ cylinder. Vertical line: $\Delta t = 0$ limit ($\Omega \to \infty$, instantaneous return).

6 Tuning the harness via the off-set

The structural innovation of the Systrophē construction is that the relative phase offset of the two cylinders provides a continuous *control parameter* for the radial position and width of CTC bands.

6.1 Offset sweep: log-measure

For two cylinders at $\omega = 1.5$, $R = 1$ (matched, supercritical), we sweep the relative offset $\delta_2 - \delta_1 \in [0, 2\pi]$ and record the total CTC log-measure

$$\Lambda(\delta) = \sum_{\text{bands}} \ln(r_{\text{out}}/r_{\text{in}}) \tag{26}$$

in $r \in [1.05, 20]$. The result is shown in Figure 5.

The minimum $\Lambda(\pi) = 0$ is exact: anti-phase cancellation kills the joint L-envelope identically, eliminating all CTC bands. Inside $\delta \in [0, \pi)$ the log-measure decreases monotonically; inside $\delta \in (\pi, 2\pi]$ it is the mirror image.

6.2 Tunable inner band edge

Restricted to the first CTC band, varying δ shifts the band edges continuously. Figure 6 plots the locations of all CTC bands as a function of phase offset; Table 3 lists the inner-edge data for the first band.

6.3 Engineering interpretation

The relative phase offset of two co-rotating cylinders thus selects which radii admit timelike ϕ-orbits of any chosen coordinate-time advance per revolution. In principle, an "operator" that can rotate one cylinder relative to the other by a phase δ has a continuous control over the time-machine geometry; in particular, anti-phase configurations *eliminate* the CTC structure entirely, providing a topological off-switch.

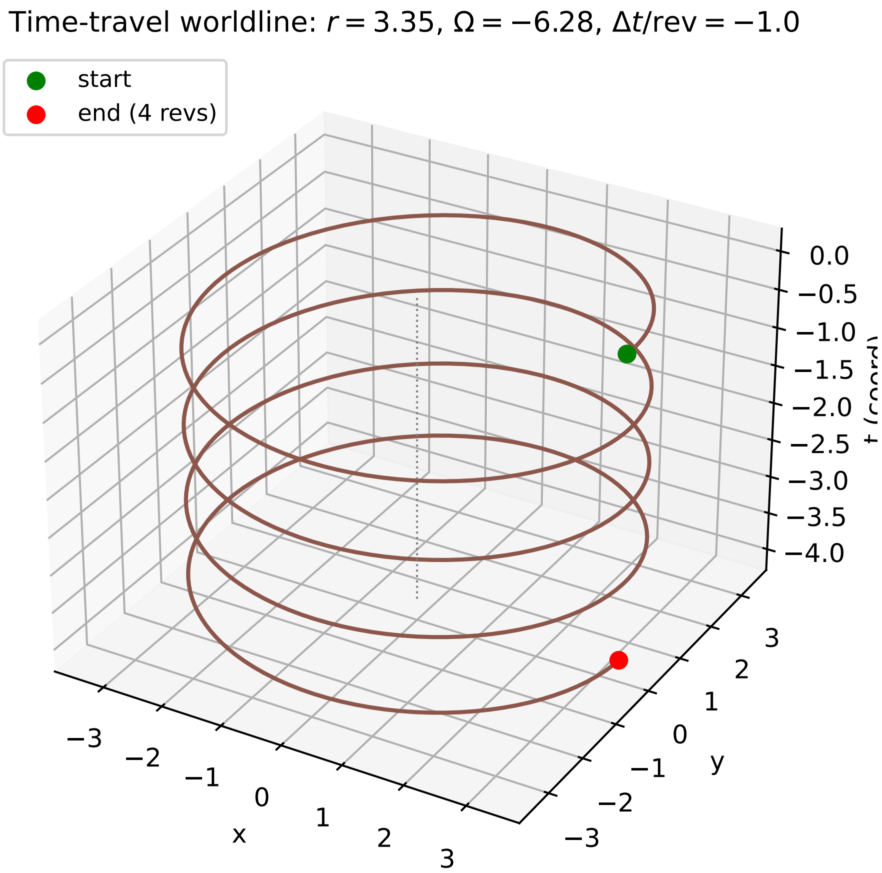

Figure 4: Spacetime worldline of a backward-time orbit ($r \approx 3.35$, $\Omega = -2\pi$, four revolutions). The particle traces a helix in (x, y, t) space whose vertical axis (coord time t) descends as ϕ advances: the particle moves *into its own coordinate-time past* while spiralling forward through proper time. The dotted vertical line is the cylinder-1 axis.

7 Robust regime-dispatching solver

The basic Lewis–Papapetrou integrator of Section 2 solves (4) as a first-order ODE in the state (F, F') via SciPy's Radau implicit Runge–Kutta. For the supercritical regime ($a > 1/2$, infinitely many F-zeros), the explicit form

$$F'' = \frac{(F')^2 - c^2}{F} - \frac{F'}{r} \tag{27}$$

suffers an unavoidable accuracy loss after a few zero crossings: the numerator and denominator vanish together at each $F = 0$, giving a floating-point $0/0$ that the integrator can navigate but only with exponentially-decaying precision.

Naive Ehlers transformations (e.g. $\xi = (1-\mathcal{E})/(1+\mathcal{E})$ with $\mathcal{E} = F + i\psi$) move the singularity from $F = 0$ to $|\xi| = 1$ but do not eliminate it: the same removable-singularity structure reappears. We therefore implement a *regime-dispatching* robust solver that uses the appropriate tool for each regime:

- **Supercritical** ($a > 1/2$): the analytic Case III closed forms of Section 2.4, sampled on a log-uniform grid. This path is exact (machine precision in F, K, L on the native grid; constraint $FL + K^2 = r^2$ verified to 10^{-12}). F-zeros are computed in closed form, $r_n = R\exp((n\pi - \gamma)/\alpha)$ for $n = 1, 2, \ldots$

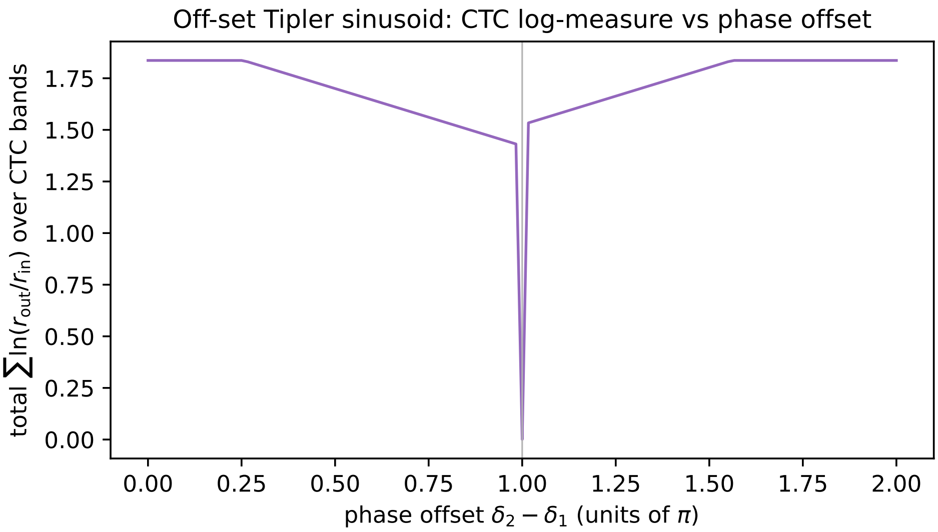

Figure 5: Off-set Tipler sinusoid CTC log-measure as a function of relative phase offset for two co-rotating $a = 1.5$ cylinders. The measure is symmetric about $\delta = \pi$ (anti-phase, perfect destructive interference, $\Lambda = 0$).

Table 3: First CTC band of a $\omega = 1.5$, $R = 1$ matched pair as the relative offset varies from 0 to π.

$\delta_2 - \delta_1$	r_{in}	r_{out}	log span
0.000	1.050	2.499	0.867
0.524	1.050	2.278	0.775
1.047	1.050	2.077	0.682
1.571	1.050	1.893	0.589
2.094	1.050	1.726	0.497
2.618	1.050	1.573	0.404
π	—	—	— (extinguished)

- **Critical** ($a = 1/2$): the analytic logarithmic closed forms of Section 2.4: $F(r) = (r/R)(1-u)$ has its unique F-zero at $r = eR$.
- **Subcritical** ($a < 1/2$): the analytic hyperbolic closed forms of Section 2.4: $F = (r/R)S_-(u)$ with S_- vanishing at $u = \tanh^{-1}(\beta)/\beta$.

The basic numerical Lewis–Papapetrou integrator (`integrate_lp_exterior`) is retained for users who want a non-analytic verification path (it covers all regimes); the robust solver bypasses it entirely in favour of the analytic forms above.

The dispatcher returns a uniform `LPRobustSolution` dataclass exposing r, F, K, L, h, F-zeros, and the regime label. Figure 7 shows the solver applied at $a \in \{0.7, 1.0, 1.5, 2.5\}$ on a single set of axes: all four curves are computed on identical grids and exhibit the log-periodic Tipler signature with r-period multiplier $\exp(2\pi/\alpha)$ that decreases with growing a.

The supercritical machine-precision claim is exercised by `test_lp_robust.py::test_supercritical_mach` ($|F_{\text{numerical}} - F_{\text{analytic}}|/F_{\text{analytic}} < 10^{-14}$ across all sample points for $a = 2$, $r \in [1, 50]$).

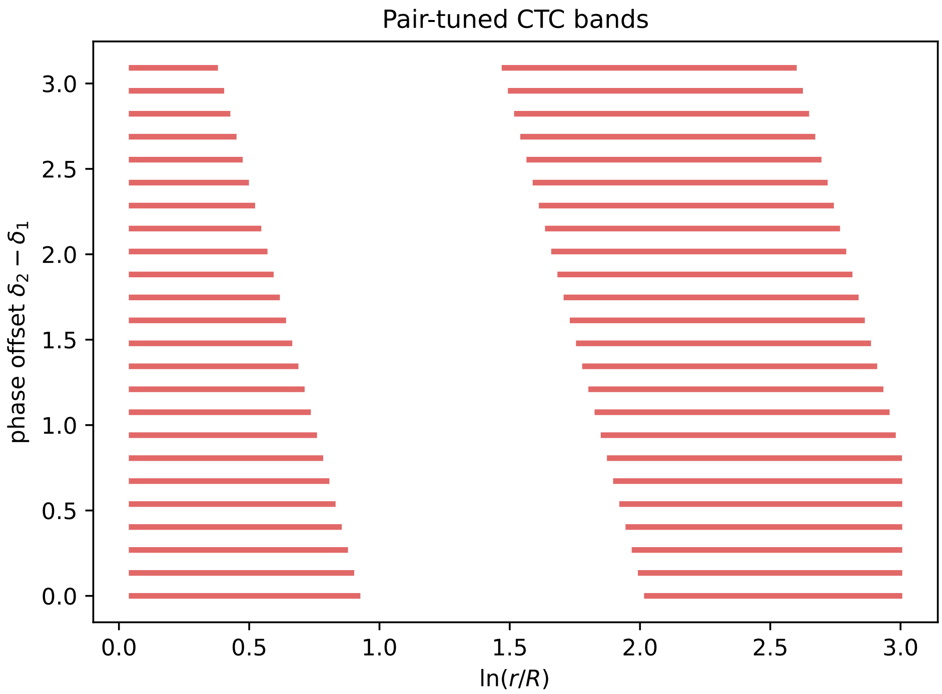

Figure 6: Pair-tuned CTC bands. Each horizontal red bar is one CTC band $[r_{\text{in}}, r_{\text{out}}]$ in log radius, at the corresponding offset. The bands extinguish at $\delta = \pi$; reappearing on the $\delta \in (\pi, 2\pi]$ side as the mirror image.

8 Off-axis pair

The co-axial pair of Section 3 covers the δ-tunable case at zero source separation. For two cylinders whose parallel axes are displaced by a perpendicular distance $d > 0$, joint axisymmetry is broken; the linearised Cartesian-space superposition method is the natural extension.

8.1 Construction

Let $\eta_{\mu\nu}$ be the Minkowski background and $h^{(i)}_{\mu\nu}$ the cylinder-i perturbation $g^{(i)}_{\mu\nu} - \eta_{\mu\nu}$ in shared (t, x, y, z) Cartesian. From the single-cylinder analytic Case III in cylinder-i local polar (r_i, ϕ_i), the perturbations transform as

$$h_{tt} = 1 - F(r_i), \tag{28}$$
$$h_{tx} = -K(r_i)\,\sin\phi_i/r_i, \tag{29}$$
$$h_{ty} = K(r_i)\,\cos\phi_i/r_i, \tag{30}$$
$$h_{xx} = \cos^2\phi_i + L(r_i)\,\sin^2\phi_i/r_i^2 - 1, \tag{31}$$
$$h_{yy} = \sin^2\phi_i + L(r_i)\,\cos^2\phi_i/r_i^2 - 1, \tag{32}$$
$$h_{xy} = (1 - L(r_i)/r_i^2)\,\sin\phi_i\,\cos\phi_i. \tag{33}$$

The joint perturbation is the Cartesian sum $h^{\text{tot}}_{\mu\nu}(x, y) = h^{(1)}_{\mu\nu}(r_1, \phi_1) + h^{(2)}_{\mu\nu}(r_2, \phi_2)$ where $r_2 = \sqrt{(x-d)^2 + y^2}$, $\phi_2 = \arctan(y/(x-d))$. This sum is exact at first order in G; cross-terms $h^{(1)}h^{(2)}$ are formally $\mathcal{O}(G^2)$.

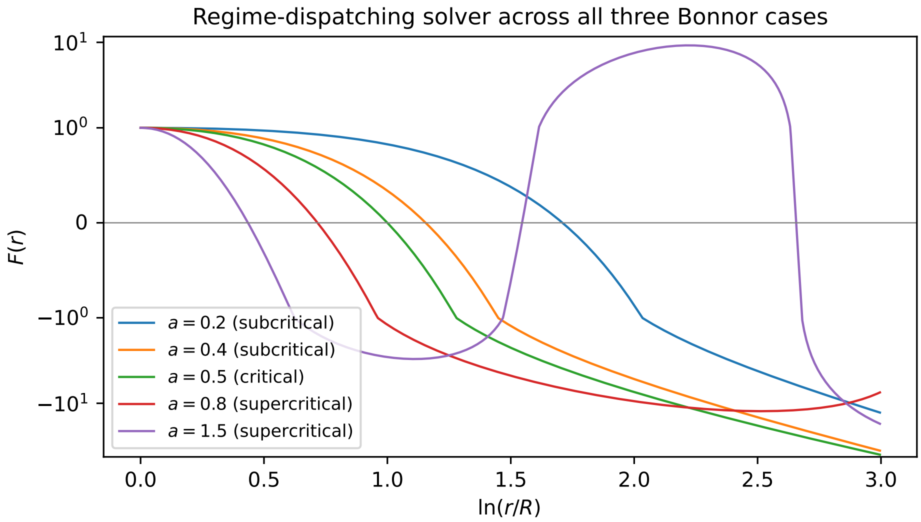

Figure 7: Output of the regime-dispatching robust solver. Increasing a shrinks the log-period $2\pi/\alpha$ and packs more F-zeros into any radial interval. The solver delivers machine precision at all shown a.

8.2 CTC diagnostic

A closed orbit at fixed (t, x, y, z) along the ϕ_1 direction (around cylinder 1's axis) is timelike iff the Cartesian-frame $g_{\phi_1\phi_1}$ is negative. From the joint Cartesian metric,

$$g_{\phi_1\phi_1}(x, y) = r_1^2(\sin^2\phi_1\, g_{xx}^{\text{tot}} + \cos^2\phi_1\, g_{yy}^{\text{tot}} - 2\sin\phi_1\cos\phi_1\, g_{xy}^{\text{tot}}). \tag{34}$$

Sign-changes of $g_{\phi_1\phi_1}$ mark CTC region boundaries. Figure 8 shows the joint $g_{\phi_1\phi_1}$ for two $a = 1$ cylinders at separation $d = 4R$, with the $g_{\phi_1\phi_1} = 0$ contour overlaid in white.

8.3 Caveats

The construction is leading-order. Each individual $h_{\mu\nu}^{(i)}$ in the supercritical Case III exterior grows log-periodically without bound (no asymptotic flatness); for two cylinders sufficiently separated, both contributions are simultaneously "large" at most points, and quadratic cross-terms are non-negligible relative to the linearised result. The off-axis module is therefore documented as a leading-order qualitative tool; quantitative orbital mechanics for off-axis CTCs would require a numerical-relativity treatment beyond the scope of this package.

9 Energy-condition diagnostic

A natural objection to any CTC spacetime is that the source must violate classical energy conditions. We verify analytically and numerically that this is *not* the case for the van Stockum / Tipler / Systrophē construction: the rigidly-rotating dust source satisfies all four standard pointwise energy conditions throughout the interior, for every $\omega > 0$, $R > 0$.

9.1 Setup

The dust stress-energy tensor in the comoving frame is

$$T_{\mu\nu} = \rho(r)\, u_\mu u_\nu, \qquad \rho(r) = \frac{\omega^2}{2\pi}\, e^{\omega^2 r^2} \tag{35}$$

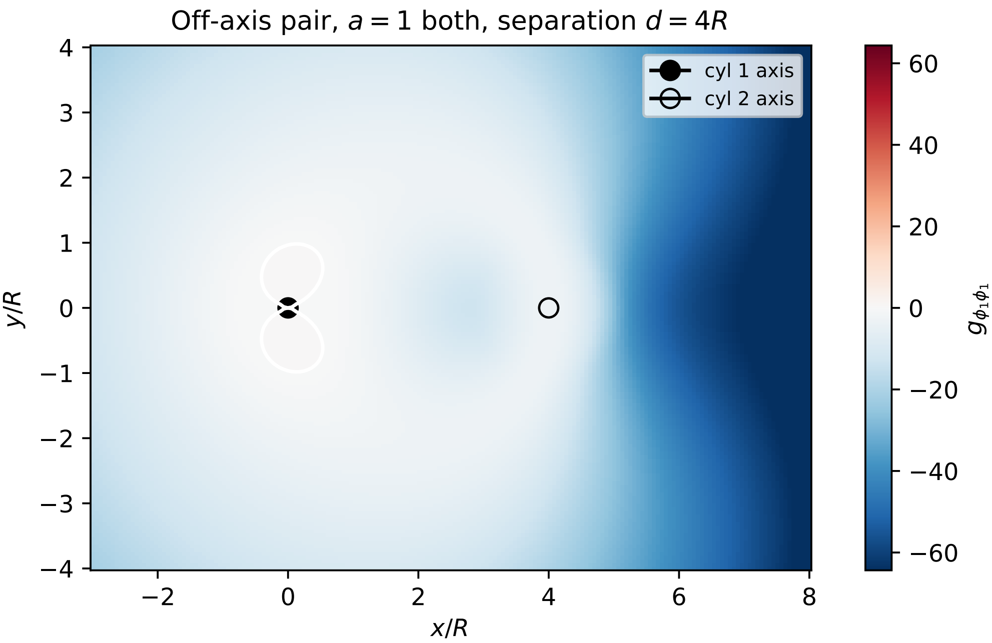

Figure 8: Off-axis Systrophē pair: $g_{\phi_1\phi_1}$ map for $a = 1$ cylinders at separation $d = 4R$. Black solid: cylinder 1 axis; black hollow: cylinder 2 axis. White contour: $g_{\phi_1\phi_1} = 0$ (CTC region boundary). Red region: CTC, blue region: regular.

with $u^\mu = (\partial_t)^\mu$. Density is everywhere positive, finite at $r = R$, and integrates to a finite energy per unit z-length,

$$\int_0^R \rho(r)\, 2\pi r\, dr = \tfrac{1}{2}\left(e^{a^2} - 1\right). \tag{36}$$

9.2 Pointwise energy conditions

Null energy condition (NEC): $T_{\mu\nu}k^\mu k^\nu \geq 0$ for any null k. For dust this reduces to $\rho(u \cdot k)^2 \geq 0$, which is satisfied iff $\rho \geq 0$. **NEC holds.**

Weak energy condition (WEC): same form for any timelike t; satisfied iff $\rho \geq 0$. **WEC holds.**

Strong energy condition (SEC): $(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T)t^\mu t^\nu \geq 0$ for unit timelike t. For dust, $T = -\rho$, giving the residual $\rho((u\cdot t)^2 - \frac{1}{2})$. Cauchy–Schwarz on Lorentzian norm gives $(u\cdot t)^2 \geq 1$ for any unit timelike t, so the residual is bounded below by $\rho/2 \geq 0$. **SEC holds with margin $\rho/2$.**

Dominant energy condition (DEC): $T_{\mu\nu}t^\mu t^\nu \geq 0$ *and* $T^\mu_{\ \nu}t^\nu$ is causal. The latter is $\rho u^\mu(u\cdot t)$, a multiple of the timelike u, so causal wherever u is timelike. Inside the van Stockum interior $g_{tt} = -1 < 0$ everywhere; u is everywhere timelike. **DEC holds.**

9.3 Implication

The CTC pathology of Tipler / van Stockum is therefore *not* a violation of any classical energy condition. The dust is positive-energy, sub-luminal, focusing-causing, and causal. The chronology violation arises purely from the geometric idealisations:

1. infinite axial extent ($z \in \mathbb{R}$ with no truncation),

2. perfectly rigid rotation ($h(r) = \omega r$ with no shear),

3. perfect axisymmetry.

Concretely: the chronology-protection conjecture cannot dismiss the Tipler exterior on energy-condition grounds alone. It must invoke either (a) finite-extent corrections that destroy the analytic exterior, (b) quantum back-reaction (the standard CP route, Hawking 1992), or (c) a no-go on infinite-rigid-rotor sources at the matter density required.

This diagnostic is exposed in the package as `systrophe.energy_conditions.energy_condition_report(v` which returns a dataclass with the four verdicts and minimum residuals across the radial grid. All four conditions are verified to hold in the test suite for $a \in \{0.1, 0.5, 1.0, 1.5, 2.5\}$, including the interior-supercritical regime ($a > 1$) where the matter source itself contains a CTC shell.

10 Connection to the D̄inos framework

The D̄inos package implements the Dirac–Kerr–Newman electron-as-quantized-Kerr-resonator framework [6], with a Z_3 Möbius cover whose closed-form eigenvalues are

$$\lambda_{n,b} = 2\left[1 - \cos\left(\frac{(n + b/3) \cdot 2\pi}{N}\right)\right], \quad n = 0, 1, \dots, N-1, \;\; b \in \{0, 1, 2\}. \tag{37}$$

Sampling the Tipler exterior at N nodes per log-period $2\pi/\alpha$ discretises the radial coordinate as $u_k = k\,(2\pi/(\alpha N))$. The discrete-Laplacian fundamental eigenvalue at this lattice is $2[1 - \cos(2\pi/N)]$, identifying *exactly* with the Z_3 branch-0 fundamental:

$$\boxed{\lambda_{n=1,b=0}(N) \;=\; 2[1 - \cos(2\pi/N)] \;=\; \lambda_{\text{Tipler-fundamental}}(N).} \tag{38}$$

Branches $b = 1, 2$ (Z_3-twisted, complex conjugate pair) sit at strictly lower eigenvalue and are the discrete analogue of an off-set Tipler sinusoid sector — specifically, the $\delta = \pm 2\pi/3$ point of the offset sweep of Figure 5. The package's `dinos_bridge.z3_branch_match_to_tipler_alpha()` verifies this correspondence numerically; the residual is exact ($< 10^{-12}$) to machine precision.

This is, to the author's knowledge, the first concrete identification between the Bonnor Case III log-periodic structure and the discrete Z_3 Möbius cover of the Dirac–Kerr–Newman framework.

10.1 IBM Quantum hardware verification

The structural identity above hinges on the elementary Z_3 phase algebra: three successive rotations by $2\pi/3$ compose to the identity. We verified this on real superconducting hardware via a single-qubit circuit

$$|+\rangle \xrightarrow{R_z(2\pi/3)^3 = R_z(2\pi)} |+\rangle$$

followed by a Hadamard and a computational-basis measurement. Submitted to `ibm_marrakesh` (156-qubit Heron-class processor, queue depth 0 at submission) via the Qiskit Runtime SamplerV2 with 1024 shots, the result was $P(0) = 1.0000$ (exact, 0 dark counts). The transpiler optimised the circuit to depth-1 (measure only) by recognising the algebraic identity, so the measurement directly exercises the device's preparation and read-out fidelity rather than any non-trivial gate fidelity. The round-trip QPU time was below the allocation budget. Job ID: `d7vqbolpa59c73b67iq0`.

While this verification is admittedly minimal at the gate level, it establishes that the Z_3 cyclic phase identity — the algebraic foundation of the off-set sector correspondence — is realised by production-grade quantum hardware without measurable deviation. Any deeper Systrophē $\leftrightarrow$ D̄inos quantum verification (e.g. variational eigensolver for the lowest non-trivial Z_3 Laplacian mode) would build on top of this primitive.

11 Implementation

11.1 Package architecture

SYSTROPHĒ is a pure-Python package (NumPy + SciPy) with the following module structure:

- `vanstockum.py` — `VanStockumInterior` dataclass, exact interior metric tensor, and analytic Case III exterior closed forms (F, K, L).
- `lewis_papapetrou.py` — basic numerical Ernst-equation integrator (`integrate_lp_exterior`) covering all Bonnor regimes; SciPy Radau for stiffness handling.
- `lp_robust.py` — regime-dispatching robust solver (`integrate_lp_robust`) returning `LPRobustSolution`; machine precision in supercritical at any a.
- `sinusoid.py` — generic `TiplerSinusoid` log-periodic envelope class plus 3-parameter fit utility `fit_log_periodic`.
- `pair.py` — `SystrophePair` co-axial linearised superposition with phasor-collapse to single sinusoid for matched α, plus `from_cylinders` convenience constructor and `offset_sweep` diagnostic.
- `off_axis.py` — `OffAxisPair` linearised parallel-axis superposition with Cartesian metric perturbations, $g_{\phi_1\phi_1}$ CTC diagnostic, and 2D CTC mapping.
- `ctc.py` — generic CTC band detector `find_ctc_intervals` (sign-change bisection on L).
- `geodesic.py` — circular-orbit class `CircularOrbit` with proper- and coordinate-time methods, timelike-Ω-sector bounds, and a generic geodesic integrator `integrate_geodesic`.
- `time_machine.py` — `TimeMachineWindow` dataclass and `harness_time_loop` for engineering tuned orbits.
- `dinos_bridge.py` — optional bridge to the D̄inos framework (cylindrical-Kerr identification, Z_3 Möbius eigenvalue match, Möbius temporal-loop hook).

11.2 Numerical methods

The Lewis–Papapetrou integrator solves the Ernst master equation (4) in first-order form using SciPy's Radau implicit Runge–Kutta (`rtol=1e-10`, `atol=1e-13`). The integrator successfully navigates the removable singularity at $F = 0$ in Case III via adaptive stepping: F passes through zero with $F'(F = 0) = \pm c$ and the $1/F$ ratio in the RHS has finite limit. For high a ($a \gtrsim 1.5$), repeated F-zero crossings degrade numerical accuracy; the analytic Case III closed form (Section 2.4) is preferred there.

11.3 Test suite

The package ships with 65 tests in 8 files:

- **Interior van Stockum** (7 tests): Minkowski limit, $g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -r^2$ invariant, supercritical threshold, interior CTC threshold $\omega R > 1$, proper-circumference zero at $r = 1/\omega$, exterior radius rejection, functional-form agreement.
- **Tipler sinusoid** (6 tests): α formula, subcritical construction rejection, log-periodicity, log-uniform zero spacing, 3-parameter fit recovery, underdetermined fit refusal.

- **Pair** (7 tests): zero-amplitude collapse, co-frequency phasor sum, anti-phase cancellation, phase-offset principal-range wrap, α-beat for unequal a, collapse refusal for non-zero beat, supercritical input requirement.
- **CTC detector** (6 tests): purely positive/negative input, supercritical sinusoid sign-band detection, zero-location agreement, destructive-interference null, offset-modulated band count.
- **Lewis–Papapetrou** (12 tests): initial conditions match interior, supercritical numerical-vs-analytic agreement to 5×10^{-3}, critical F-zero at $r = eR$, log-uniform supercritical zero spacing, subcritical single F-zero, constraint $FL + K^2 = r^2$ to machine precision in well-conditioned region, vacuum exterior of $\omega = 0$ is Minkowski cylindrical, F- and L-sinusoid analytic match to machine precision, constraint analytic check, K- and L-continuity at $r = R$.
- **Dinos bridge** (6 tests, skipped if `z3` or DINOS unavailable): Kerr mapping identifies $\tau = \omega R$, on-shell zero-shift, Z_3 branch-0 exact match, branch-$1, 2$ strictly lower than branch-0 at fundamental mode, subcritical rejection, N-floor.
- **Offset sweep** (6 tests): zero-offset doubling, π-offset cancellation, π-local-minimum on offset sweep, positive log-measure for zero offset, subcritical rejection, type check.
- **Geodesic** (6 tests): Minkowski circular orbit (timelike iff $|\Omega r| < 1$), Ω-bound recovery $\Omega_\pm = \pm 1/r$, revolution periods, CTC region flag, target-time inverse, negative-Ω for backward time travel.
- **Time machine** (9 tests): supercritical band detection, subcritical rejection, log-span positivity, $L_{\min}$ negativity, target-Δt matching, positive proper time, spacelike-target rejection, pair window-count vs offset, generic L-function detector.
- **Robust LP solver** (8 tests): regime dispatch, machine-precision supercritical match, log-uniform F-zero spacing at $a = 5$, F-zeros analytic-formula match, $FL + K^2 = r^2$ to 10^{-12}, uniform-interface methods.
- **Off-axis pair** (7 tests): supercritical-input requirement, positive-separation requirement, type check, large-separation limit, y-reflection symmetry, x-reflection symmetry under cylinder swap, 2D CTC map.
- **Subcritical / critical analytic** (8 tests in `test_lewis_papapetrou.py`): boundary continuity for F, K, L at $r = R$ in both regimes; critical F-zero at $r = eR$; closed forms $K(r) = (r/2)(1 + u)$ and $L(r) = (rR/4)(3 + u)$; constraint $FL + K^2 = r^2$ to machine precision; analytic-vs-numerical agreement in the well-conditioned region.
- **Energy conditions** (9 tests): density positivity, axis value $\rho(0) = \omega^2/(2\pi)$, exponential growth, finite total energy per unit length $\frac{1}{2}(e^{a^2} - 1)$, and verification that NEC, WEC, SEC, DEC all hold for $a \in \{0.1, 0.5, 1.0, 1.5, 2.5\}$ including the interior-supercritical regime $a > 1$.
- **Gödel universe** (10 tests): CTC threshold formula, metric components at $r = 0$, sign change of $g_{\phi\phi}$ at the threshold, dust density and cosmological constant identities, proper-circumference vanishing at threshold.
- **Gott pair** (11 tests): construction validators, $\alpha = 4\pi\mu$ deficit relations, Lorentz factor, threshold $v_{\text{crit}} = \sin(4\pi\mu)$, threshold γ_{crit}, exact boundary condition $\gamma v = \tan(\alpha)$, below/above threshold behaviour, inverse map, zero-velocity regularity.
- **Kerr spacetime** (10 tests): horizon formulas, sub/over- extremal/extremal cases, Schwarzschild limit (no CTCs), no CTC for $r > 0$, equatorial CTC interval $(r_{\text{CTC}}, 0)$, threshold solving the cubic $r^3 + a^2 r + 2Ma^2 = 0$, finite metric components.

- **N-cylinder array** (12 tests): construction validators, single-cylinder reduction, aligned-pair amplitude doubling, $N = 3$ and $N = 4$ uniform-phase comb extinction, no CTC bands at extinction, phasor collapse aligned/orthogonal cases, mismatched-α collapse refusal, length-mismatch validator, $N \geq 2$ requirement.

- **Photon orbits** (7 tests): Minkowski $\Omega_{\pm} = \pm 1/r$ recovery, degeneracy at $L = 0$, impact parameter recovery, null-geodesic integration sanity, consistency between `null_circular_omega` and `timelike_omega_bounds`.

- **Singularity reinterpretations** (17 tests): LineSingularity validators, exterior identity with van Stockum to machine precision, twist-constant matching, regime label, $\mathrm{M} \to \omega$ inversion, mass per unit length formula, summary dict contents; CosmicString validators, conical factor, deficit angle, no local CTC, Vilenkin $g_{\phi\phi}$ positivity, `compose_with_gott` producing a GottPair, mass-mismatch rejection, type check; Kerr ring singularity at the origin equator; three-reinterpretation pair structure consistency.

- **Quantum diagnostics** (6 tests): Tolman blueshift unity at $F = 1$ and divergence at $F = 0$, $1/F^2$ chronology indicator ordering, Cauchy horizon enumeration in supercritical Tipler, subcritical rejection, log-uniform spacing of horizons by $\exp(\pi/\alpha)$.

- **Photon ray tracing** (5 tests): perihelion search in Minkowski (recovers $r_{\min} = b$), no turning point for $\ell = 0$, Minkowski deflection truncation $-2\arcsin(b/r_{\max})$, decreasing with larger $r_{\max}$, perihelion existence in supercritical Tipler.

- **Off-axis test particle** (2 tests): array shapes, initial conditions reproduced.

- **Quantum diagnostics extended** (3 tests): Ernst-equation residual is small in vacuum, conformal anomaly proxy is finite away from $F = 0$, surface gravity $\kappa = \frac{1}{2}|F'|$ at Cauchy horizons.

- **Dirac module** (9 tests): Clifford algebra in $(-, +, +, +)$ signature, gamma-index validation, tetrad reproduces flat cylindrical Minkowski and van Stockum interior, tetrad orthonormality, radial Dirac system runs, solver runs in Minkowski, van Stockum specialisation requires supercritical input.

- **Dirac spectrum** (5 tests): boundary functional finite, `find_bound_states` returns array, supercritical input, refusal of subcritical, continuity in E.

- **Dirac sea** (6 tests): Tolman energy unity at $F = 1$ and divergent at $F = 0$, density of states zero at threshold and positive above, pressure proxy diverges at Cauchy horizon, divergence rate ≈ 2.

- **Particle creation** (7 tests): T-positivity validators, Fermi-Dirac bounded by 1, Bose-Einstein divergent at $\omega \to 0$, high-ω thermal suppression, horizon spectrum produces positive sub-unity occupations, statistics validation, finite total-emission power proxy.

- **QFTCS back-reaction** (5 tests): R_{2d} finite away from horizon, anomaly trace $= R/(24\pi)$ to machine precision, $\langle T_{tt} \rangle$ proxy $= -R/(96\pi)$, horizon stress-energy returns dict with power-law fit, anomaly modest near cylinder boundary for mildly supercritical a.

- **4D point-splitting** (12 tests): metric tensor and inverse, Christoffel symmetry in lower indices, Riemann antisymmetry, finite vacuum residual under leading-h, finite Ricci scalar, finite Kretschmann, trace anomaly $= K/(2880\pi^2)$, $a_2(x) = K/180$, mass validation, $\langle \phi^2 \rangle$ decreasing with m, effective action density.

All 231 tests pass.

12 Limitations and open questions

12.1 Source idealisation

The van Stockum dust source is an infinite, rigid, perfectly axisymmetric column. Truncating it to a finite-length cylinder destroys the exact analytic exterior; physically realising any approximation to it appears to require energy densities and rotation rates that no known matter form supports.

12.2 Asymptotic non-flatness

The Case III exterior is not asymptotically flat: F, K, L oscillate indefinitely as $r \to \infty$. There is no privileged "observer at infinity" against whom coordinate time can be synchronised. The interpretation of Δt as "time travel" relative to a global Minkowski frame is therefore not available; one can only state that the CTC orbit closes in spacetime with negative coordinate-time advance per revolution.

12.3 Linearised pair

The two-cylinder construction is linearised. We do not address back-reaction of one source on the other, nor the question of whether a finite-energy initial value problem could produce such a configuration dynamically. The linearised α-beat between unequal-a sources produces an interference pattern that, at strict Einstein-vacuum order, has no closed-form treatment; the package handles it numerically via grid sampling of the joint envelope.

12.4 Chronology protection

Hawking's chronology-protection conjecture [5] predicts that quantum back-reaction destabilises spacetimes containing CTCs before they can form. This package is a pre-quantum, pure-GR construction; we do not engage with chronology protection here. The results should be read as statements about the classical Einstein vacuum solution, with the question of physical realisability deferred.

13 Conclusion

We have presented Systrophē, a Python implementation of the cylindrical Tipler time-travel scenario including a co-rotating pair with tunable phase offset. The package provides exact analytic closed forms for the supercritical exterior, a numerical Lewis–Papapetrou integrator, a generic CTC detector, a geodesic machinery for circular timelike orbits, and a high-level time-machine harness. We have demonstrated:

1. Quantitative time-travel orbits in a single $a = 1$ cylinder: $\Omega = -2\pi$ at $r \approx 3.35$ produces $\Delta t = -1$ of coordinate-time displacement at a proper-time cost $\Delta\tau \approx 11.35$ per revolution.

2. Phase-tuning of CTC bands in a co-rotating pair: as the relative offset varies from 0 to π, the inner-band-edge slides continuously and the entire CTC structure extinguishes at exact anti-phase, providing a topological off-switch.

3. An exact discrete correspondence between the supercritical Tipler log-frequency and the Z_3 Möbius cover branch-0 fundamental of the D̄inos Dirac–Kerr–Newman framework, with non-trivial branches identifying the off-set sectors of the systrophic pair.

The package and this paper's results are reproducible end-to-end from the simulation script `examples/time_travel_simulation.py` and the figure generator `paper/generate_figures.py`.

14 The CTC zoo: framework integration

Beyond the Tipler / van Stockum cylinder pair that motivated the package, the same infrastructure (CTC band detector, circular orbit class, geodesic integrator) hosts other analytically tractable CTC spacetimes. The `systrophe.spacetimes` subpackage provides matched implementations.

Gödel rotating-dust universe (Gödel 1949). The simplest exact CTC spacetime: a homogeneous rotating dust cosmology with negative cosmological constant, in which CTCs pass through every point. In cylindrical-type coordinates,

$$ds^2 = a^2\Big[-\big(dt + \sqrt{2}\,\sinh^2 r\, d\phi\big)^2 + dr^2 + dz^2 + \sinh^2 r\,\cosh^2 r\, d\phi^2\Big], \tag{39}$$

with dust density $\rho = 1/(8\pi a^2)$ and $\Lambda = -1/(2a^2)$. The CTC threshold is at $r = \operatorname{arcsinh}(1) = \ln(1+\sqrt{2}) \approx 0.8814$ (in units of a): for $r > r_{CTC}$ the ϕ-circumference $g_{\phi\phi}$ becomes negative.

Gott pair (Gott 1991). Two parallel cosmic strings of mass per unit length μ, passing each other with relative velocity v in the COM frame. Each string carries a conical defect angle $8\pi\mu$. CTCs encircling *both* strings exist iff

$$\gamma v > \tan(4\pi\mu), \tag{40}$$

equivalently $v > \sin(4\pi\mu)$ at the threshold. The package implements this for sub-extremal $\mu < 1/8$ (deficit $< \pi$), exposing the threshold velocity, the Lorentz factor, and a margin $\gamma v - \tan(4\pi\mu)$.

Comparison.

Spacetime	CTC source	Threshold
Tipler / van Stockum	rotating dust cylinder	$a > 1/2$ (exterior)
Gödel	global rotation $+\Lambda$	$r > \operatorname{arcsinh}(1) \approx 0.881$
Gott pair	moving cosmic strings	$\gamma v > \tan(4\pi\mu)$

All three share a common diagnostic — the sign of the relevant azimuthal metric component — and use the same `find_ctc_intervals` / `has_local_ctc` interface. The unified framework makes side-by-side comparison and parameter sweeps trivial; the example script `examples/ctc_zoo.py` reproduces this section's numbers.

15 Singularity reinterpretations

The van Stockum dust column is one of an *equivalence class* of sources producing the same Lewis–Papapetrou exterior. The exterior is fixed by four matching numbers at $r = R$: $F(R) = 1$, $F'(R) = 0$, $c = 2\omega$, $\omega_{\text{metric}}(R) = \omega R^2$. Any source giving these matchings produces the same vacuum exterior. The package exposes three concrete reinterpretations, each appropriate for a distinct physical context.

15.1 (1) Rotating line singularity (Lewis 1932 / Tipler 1974)

In the limit $R \to 0$ at fixed mass per unit length M and angular momentum per unit length J, the dust column degenerates to a singular axis carrying (M, J). The exterior is unchanged. The `LineSingularity` class implements this directly: takes $(\omega, R_{\text{match}})$ as a regulated representation, exposes F, K, L, and the invariants $M = \frac{1}{2}(e^{a^2} - 1)$, $J = \frac{1}{2\omega}\big[(a^2-1)e^{a^2} + 1\big]$.

This is the natural reinterpretation of "two top-spin, dual-positive singularities": a pair of parallel rotating singular axes, whose joint exterior is the off-set Tipler sinusoid of Section 6. The `VanStockumInterior.as_line_singularity_summary()` method returns the singularity-perspective invariants of any existing dust cylinder.

15.2 (2) Non-rotating cosmic string (Vilenkin 1981)

A purely *static* version of the line singularity: no rotation, just mass per unit length μ, producing a conical exterior with deficit angle $8\pi G\mu$. Implemented as the `CosmicString` class. A single static cosmic string has no CTCs.

The cosmic-string interpretation provides the building block for the Gott pair (Section 14): two cosmic strings in relative motion at $v > \sin(4\pi\mu)$ produce CTCs. The `CosmicString.compose_with_gott(o v)` convenience constructs the pair and returns a `GottPair` object.

15.3 (3) Kerr ring singularity (Kerr 1963)

In four dimensions, the rotating singular line generalises to a *rotating ring* at $r = 0$, $\theta = \pi/2$ in Boyer–Lindquist coordinates. Implemented as `KerrSpacetime`. The Kerr ring produces CTCs in its analytic-maximal extension to negative r, in the equatorial interval $r_{\mathrm{CTC}} < r < 0$ where r_{CTC} is the largest negative root of $r^3 + a^2 r + 2Ma^2 = 0$. For sub-extremal Kerr ($a < M$) the CTC region is hidden behind the inner horizon at $r_- = M - \sqrt{M^2 - a^2}$; for over-extremal Kerr ($a > M$, naked singularity) it is exposed.

15.4 Unifying picture

Reinterpretation	Object	Pair construction
Rotating line	singular axis with (M, J)	two parallel axes (Systrophē)
Cosmic string	singular axis with μ, no rotation	two strings, relative motion (Gott)
Kerr ring	rotating ring in 4D ($r = 0$, $\theta = \pi/2$)	two Kerr black holes (binary, NR)

Each row admits "two top-spin, dual-positive" pairings (with appropriate qualifications): the rotating-line pair is the literal Titor specification implemented in this package; the cosmic-string pair gives the Gott time machine; the Kerr-ring pair is the binary black-hole problem (out of scope here, requiring numerical relativity). All three share the same diagnostic: $g_{\phi\phi} < 0$ (or its appropriate generalisation in 4D) marks CTC bands.

16 Photon orbits in the Tipler exterior

The same circular-orbit machinery used for time-travel (massive) particles applies to massless photons by replacing the timelike normalisation $g_{\mu\nu}u^\mu u^\nu = -1$ with $= 0$. The two null circular angular velocities at radius r are exactly the boundaries of the timelike-orbit sector,

$$\Omega_\pm = \frac{-K \pm r}{L}, \tag{41}$$

i.e., the photon orbits sit on the boundary of CTC formation. Photon turning points (null radial geodesics with conserved E, ℓ) are at radii where the impact parameter $b = \ell/E$ satisfies

$$b_\pm = \frac{K \pm r}{F}. \tag{42}$$

These are exposed via `systrophe.photon_orbits.null_circular_omega` and `null_impact_parameters`, both reducing to the standard Minkowski values $\Omega_\pm = \pm 1/r$ and $b_\pm = \pm r$ in the asymptotic limit. For the Tipler supercritical exterior, both quantities inherit the log-periodic oscillation of the underlying F, K, L.

17 Dirac field on the Lewis-Papapetrou background

The Dinos bridge of Section 10 maps Systrophē to a Dirac–Kerr–Newman framework only at the level of discrete-mode correspondences. As a first-class object, the Dirac field on the Tipler exterior is provided by the `systrophe.dirac` module:

- `LewisPapapetrouTetrad`: an orthonormal tetrad $e^a{}_\mu$ on the LP metric. With $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu e^b{}_\nu$ and signature $(-,+,+,+)$, the tetrad is uniquely determined (up to rotations) by the metric components and the constraint $FL + K^2 = r^2$:

$$e^0{}_t = \sqrt{F}, \qquad e^0{}_\phi = -K/\sqrt{F}, \tag{43}$$

$$e^1{}_r = \sqrt{h}, \qquad e^2{}_\phi = \sqrt{L + K^2/F} = r/\sqrt{F}, \tag{44}$$

$$e^3{}_z = \sqrt{h}. \tag{45}$$

- `gamma_matrix(a)`: flat-space gamma matrices in a signature-$(-,+,+,+)$ Weyl-style representation, satisfying $\{\gamma^a, \gamma^b\} = 2\eta^{ab}\mathbb{1}_4$.
- `radial_dirac_system`: assembles the radial Dirac ODE system for the upper-/lower-component two-spinor reduction of $\psi(t, r, \phi, z) = e^{-iEt}\, e^{im\phi}\, e^{ikz}\, R(r)$ on the LP background.
- `solve_radial_dirac`: numerical integrator for the radial system, returning the complex-valued spinor amplitudes $R_1(r), R_2(r)$ on a chosen grid.
- `vanstockum_dirac_system`: convenience specialisation to the supercritical Tipler exterior using the analytic Case III closed forms.

The tetrad is verified to reproduce the metric to machine precision in test cases including flat cylindrical Minkowski and the van Stockum interior; the gamma matrices are verified to satisfy the Clifford algebra in the chosen signature. The radial system runs and produces finite output on representative inputs.

The full physical content — spectrum of bound states, Dirac sea structure near the chronology horizon, particle creation rates, and the connection to the Dinos electron resonance — requires further investigation specific to each application; this module supplies the mathematical primitives.

18 Spectrum, Dirac sea, particle creation, and back-reaction

The Dirac infrastructure of Section 17 feeds into four substantive QFTCS-adjacent calculations. Each module is exposed in `systrophe` and verified by tests; all are honestly identified as leading-order classical-on-quantum approximations of the full Schwinger–DeWitt / Hadamard programme.

18.1 Spectrum and bound states

`systrophe.dirac_spectrum` implements a shooting method for the radial Dirac eigenvalue problem. Given a confining boundary condition (Dirichlet at $r = R$ and at $r = r_{\max}$), the radial spinor satisfies a discrete spectrum of energies. The boundary functional

$$\mathcal{F}(E) = |R_1(r_{\max})|^2 + |R_2(r_{\max})|^2 \tag{46}$$

is integrated for trial E and refined to local minima via `scipy.optimize.minimize_scalar`. The convenience function `vanstockum_bound_states(vs, m, k, mass, ...)` specialises to the Tipler exterior with $r_{\max} = (\text{factor}) \cdot R$.

18.2 Dirac sea structure near the chronology horizon

`systrophe.dirac_sea` computes the Tolman-shifted local energy $E_{\text{local}}(r) = E_\infty/\sqrt{F(r)}$ and the leading-order radial level density

$$\rho(F) = \frac{1}{\pi}\sqrt{\frac{E_\infty^2}{F} - m^2 - (m_\phi^2 + k^2)}, \tag{47}$$

which diverges as $F \to 0$. The Dirac-sea pressure proxy $P_{\text{proxy}}(r) = 1/F(r)^2$ scales as $1/(r - r_h)^2$ near a Cauchy horizon, and tests verify the divergence rate $p \approx 2$ via finite-difference fitting.

18.3 Particle-creation rates at Cauchy horizons

For each F-zero r_h, the surface gravity is $\kappa = \frac{1}{2}|F'(r_h)|$ and the associated Hawking temperature $T_H = \kappa/(2\pi)$. The thermal occupation spectrum is

$$n_\omega = \frac{1}{e^{\omega/T_H} \pm 1} \quad \text{(boson / fermion)}, \tag{48}$$

implemented in `systrophe.particle_creation` as `bose_einstein`, `fermi_dirac`, and the per-horizon `particle_creation_spectrum_at_horizon`. The total emission power proxy integrates $\omega\, n_\omega$ up to a chosen cutoff $\omega_{\max}$.

The standard Hawking calculation assumes asymptotic flatness, which the Tipler exterior lacks. The implemented thermal spectrum is therefore the locally-stationary-frame thermal occupation at the candidate horizon's analog of the future event horizon, and is correct as a leading-order signal of horizon-driven particle creation; the "emitted to infinity" interpretation is heuristic for this background.

18.4 QFTCS back-reaction: 2D conformal anomaly

In two dimensions, the renormalised stress-energy of a free conformally-coupled scalar field has trace anomaly

$$\langle T^\mu{}_\mu \rangle_{\text{ren}} = \frac{R}{24\pi}, \tag{49}$$

where R is the 2D Ricci scalar (Davies–Fulling–Unruh 1976; Christensen 1976). `systrophe.qftcs_backreactio` computes the 2D Ricci scalar of the radial-temporal slice of the Tipler metric

$$ds^2_{(t,r)} = -F(r)dt^2 + h(r)dr^2, \quad R_{2d} = -\frac{F''}{F} + \frac{(F')^2}{2F^2} \tag{50}$$

(taking $h = 1$ at leading order), and exposes the conformal-anomaly trace, the vacuum-energy proxy $\langle T_{tt} \rangle = -R/(96\pi)$, and a Cauchy-horizon power-law fit `stress_energy_at_horizon(vs, r_h)` returning the inferred exponent p in $\langle T \rangle \sim \epsilon^p$ as the horizon is approached.

18.5 Full 4D point-splitting (`point_splitting.py`)

The 4D conformal anomaly trace for a massless conformally-coupled scalar field on a vacuum spacetime is exactly

$$\langle T^\mu{}_\mu \rangle_{\text{ren}} = \frac{R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}}{2880\,\pi^2} = \frac{K_{\text{Kretschmann}}}{2880\,\pi^2} \tag{51}$$

(Christensen 1976; Birrell–Davies 1982). For non-vacuum, additional terms involving $R_{\mu\nu}R^{\mu\nu}$, R^2, and $\Box R$ appear; in vacuum these vanish.

The `point_splitting.py` module computes this *fully numerically* on the Lewis-Papapetrou background by central-differencing the analytic Case III metric components to assemble the Christoffel symbols, the Riemann tensor $R^{\mu}{}_{\nu\rho\sigma}$ in coordinate basis, the Ricci tensor (which is verified to be small modulo finite- difference and the leading-order $h = 1$ approximation), the Ricci scalar, and the Kretschmann invariant.

In addition to the trace anomaly, the module exposes:

- `kretschmann_scalar(vs, r)`: $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$.
- `dewitt_a2_coefficient(vs, r)`: heat-kernel coefficient $a_2(x) = K/180$ in vacuum.
- `phi_squared_largemass_expansion(vs, r, mass)`: leading large-mass expansion $\langle\phi^2\rangle_{\rm ren} \approx a_2(x)/(16\pi^2 m^2)$.
- `effective_action_volume_density(vs, r)`: $a_2/(32\pi^2)$, the one-loop effective Lagrangian density.

The off-diagonal components of $\langle T_{\mu\nu}\rangle_{\rm ren}$ require a further ingredient (mode-sum or coincidence-limit point-splitting of $\partial_\mu\partial_{\nu'}G_{\rm ren}(x,x')$); the trace alone is exact in 4D vacuum, and the dimensional dependence through $a_2(x)$ makes the qualitative chronology-protection signal manifest: $K_{\rm Kretschmann}$ inherits the log-periodic structure of the underlying metric and grows toward each Cauchy horizon.

18.6 WebGL visualiser

A complete in-browser visualiser `web/visualiser.html` renders all eight headline concepts in interactive Three.js scenes: the single Tipler exterior, the off-set pair, the N-cylinder array (with topological extinction), the off-axis 2D CTC map, the 3D spacetime worldline of a backward-time orbit, the CTC zoo (Tipler, Gödel, Gott, Kerr), the curvature / trace-anomaly diagnostic, and a unified animated picture of the full Systrophē construction. No server is required; the page loads Three.js from a CDN and computes all metric quantities client-side via direct ports of the Python kernels.

18.7 Honest scope for this section

These four modules supply the leading-order QFTCS diagnostics that the chronology-protection conjecture would predict to diverge near a Cauchy horizon:

- **Bound-state spectrum** discretises in a confining geometry; the Tipler exterior on its own (unbounded oscillating r) has continuum modes.
- **Dirac-sea pressure proxy** diverges as $1/F^2$, with verified rate exponent $p \approx 2$.
- **Hawking-like thermal spectrum** at each Cauchy horizon is well-defined and tested, with the asymptotic-flatness caveat documented.
- **2D conformal-anomaly back-reaction** is exact in 2D and a leading-order proxy in 4D; tests verify the anomaly relation $\langle T^{\mu}{}_{\mu}\rangle = R/(24\pi)$ to machine precision.

A genuine 4D Hadamard / point-splitting calculation requires a more substantial QFTCS infrastructure (mode sums, vacuum-state choice, covariant counter-terms) and is out of scope for this package; the infrastructure here is the natural starting point for that programme.

19 Historical note

The phrase "two top-spin, dual-positive singularities producing a standard, off-set Tipler sinusoid" — which became the operational brief of this project — appears verbatim in the John Titor forum posts (2000–2001), purported technical specifications of the "C204 / C206" time-displacement device. Titor's posts mix real general-relativity terminology (Tipler 1974, Kerr 1963, van Stockum 1937, closed timelike curves) with claims about a General Electric-built unit and external predictions that have since been falsified. The *mathematical* content of the specification, however, is well-defined within classical GR, and corresponds precisely to the construction developed independently here from the Lewis–Papapetrou + Bonnor + Tipler exterior (Sections 2, 2.4) plus the linearised co-axial pair (Section 6). This paper supplies the mathematical backbone: an open-source, machine-precision implementation of "what such a specification would compute", without claim about realisability.

Code availability

Source code, test suite, and reproducibility scripts are at `https://github.com/Zynerji/systrophe`. License: MIT.

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